

Supporting information

The Cyanotype Process and its Potential in Chemistry Education

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Making cyanotypes – the blue photographs

A laboratory experiment for high-school students: ISCED 3 level student handout

Course: _____

Name: _____

Instructor: _____

Date: _____

Context:

Some compounds undergo chemical changes when exposed to light, which makes them useful in the process of analogue photography. One of the compounds is potassium ferrioxalate, a bright-green complex iron salt, easily prepared from basic chemicals. The process of making photographs with iron salt, the resulting image made of Prussian blue (Iron(II) ferricyanide) is typically called **cyanotype** and is one of the very first photographic processes. During this lesson, potassium ferrioxalate will be synthesised and further used in making the cyanotypes of plants.

Making of cyanotypes

Equipment: 3 beakers (100 ml), marker, tripod, wire gauze, Bunsen burner (or a hot plate), Büchner funnel, Büchner flask (250 ml), stirring rod, filtration paper, clip-in frame, matches, brush, scissors

Chemicals: students prepare solutions and mark the beakers according to the table S1:

Table S1. Preparation of the solutions

Beaker	A	B	C
Water	15 ml	45 ml	10 ml
Substance	1,1 g of iron(III) sulphate	1,1 g of potassium oxalate monohydrate	1,4 g of potassium ferricyanide
	0,76 g of oxalic acid dihydrate		

Diagram:

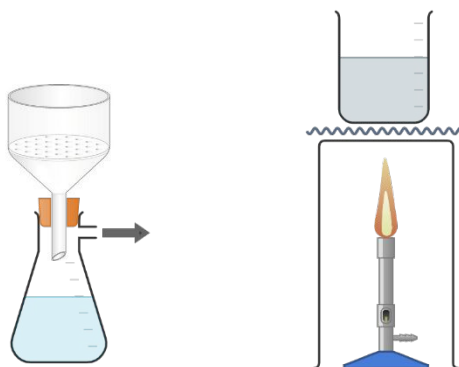


Figure S1. The diagram of the synthesis

Experimental procedure:*Step A: Potassium ferrioxalate synthesis*

1. Mark the solutions A, B, C and prepare the solutions according to the table.
2. Ask the teacher to add 5 ml of sodium hydroxide solution (20 %) into beaker A. Note the changes.
3. Filter the precipitate on the Büchner funnel.
4. Start heating beaker B and add the precipitate while mixing.
5. When all the precipitate dissolves, stop the heating and let the solution cool down.

Step B: Making of the cyanotypes

1. Mix the solutions B and C.
2. Spread the mixed solutions onto the paper using a brush.
3. Arrange objects on the paper and put it into the clip-in frame.
4. Take the frames outside and expose them to sunlight for 4 minutes. The paper should be dark blue after the exposure.
5. Wash the paper under running water until yellow areas become white.

Questions:

1. Name the brown precipitate obtained in steps A2 – 3.
2. What is the difference between artificial light and sunlight? What component of sunlight caused the sensitized paper to undergo the reaction?
3. What is the light-sensitive compound used in conventional B&W photography?